

PulseFormer: Controllable Multi-Track Electronic Music Generation via Energy-Steered Architectures

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Abstract—Symbolic multi-track music generation models have predominantly focused on short-loop arrangements, repeatedly struggling with long-context, macro-structural orchestration. Within Electronic Dance Music (EDM) composition, the interplay between dense rhythmic cohesion (Groove) and evolving dynamic intensity (Energy) constitutes the fundamental compositional backbone. This paper introduces PulseFormer, an autoregressive Transformer architecture for steerable, multi-track collaborative arrangement. We propose two architectural innovations: (1) A *Time-Major Interleaved 5D-Token* encoding that places all concurrent instruments within the same narrow token block, topologically eliminating the cross-track synchronisation burden that Track-Major architectures must bridge through long-range attention; and (2) An *Energy & Structural Indicator* pipeline enabling conditional generation that reliably obeys prompted intensity contours across multi-minute compositions. Quantitative evaluations confirm that PulseFormer provides absolute steerability—scaling from sparse introductory sections (avg. 6.9 notes/bar) to heavily-layered climax drops (surpassing 46 notes/bar)—while the Time-Major topology makes cross-track onset misalignment representationally impossible by construction.

Index Terms—Symbolic Music Generation, Transformer, Multi-track Composition, Controllability, Energy Conditioning, Electronic Dance Music

I. INTRODUCTION

The rapid proliferation of deep generative architectures has revolutionized the landscape of artificial creativity, leading to paradigm-shifting milestones in both visual synthesis and audio waveform generation. While modern audio-domain foundation models possess exceptional timbral realism and acoustic fidelity, symbolic-domain music generation (e.g., MIDI sequencing) remains fundamentally irreplaceable for contemporary professional production. Symbolic frameworks afford human composers deep structural transparency and the granular multi-track modifiability inherently required for professional mixing [1].

However, current symbolic generative models exhibit a critical limitation frequently termed the “contextual wandering” problem. When predicting discrete tokens over thousands of steps, foundational autoregressive architectures repeatedly struggle to maintain long-term macro-structural orchestration. Within highly stylized and rhythmically dense genres such as Electronic Dance Music (EDM) or modern Pop, compositional value relies less on isolated complex melodic solos, and far more on two interdependent pillars: **macro-structural energy**

evolution (i.e., the dynamic tension-and-release trajectory from a sparse 8-bar Intro cascading into a heavily saturated climactic Drop) and strict **multi-track collaborative timing** (such as the rigorous grid synchronization between a Kick drum and a Synthesizer Bassline) [2].

Conventional symbolic benchmarks unfortunately sequence multi-track data using independent, Track-Major concatenations—spooling all piano notes before proceeding to drum events. This formulation drastically severs the temporal proximity between distinct instruments, forcing the attention mechanism to reconcile massive mathematical latency windows merely to align two simultaneous rhythmic transients. Furthermore, lacking deterministic macro-orchestration parameters, most unconstrained sequence-to-sequence models naturally meander, losing the global artistic intent and deteriorating into homogenous loops devoid of narrative progression.

To solve these compounding bottlenecks, this paper introduces **PulseFormer**, an entirely novel framework pivoting strictly around structural agency and instantaneous instrumental cohesion. Rather than generating aimless symbolic streams, PulseFormer provides a mathematically steerable interface directly analogous to digital music production paradigms.

The principal contributions of this paper are summarized as follows:

- 1) **Energy-Conditioned Steerability**: We introduce explicit global prompting constraints featuring hierarchical [ENERGY_LEVEL] and [STRUCT] tags, enabling producers to literally blueprint polyphonic density waves across multi-minute audio structures deterministically.
- 2) **Time-Major Collaborative Architecture**: We propose a 5D-vector token space ordered in absolute chronological sequence across the full arrangement. By co-locating all concurrent instruments within the same narrow token window (≤ 12 tokens apart), the encoding **topologically eliminates** the cross-track synchronisation burden that Track-Major architectures must resolve through long-range attention—reducing mean co-event token distance from $\approx 3,268$ tokens to 5.2 tokens without any additional learning pressure.
- 3) **DAW-Native Sectional Stitching Logic**: An automated batch anchoring mechanism that grafts distinct generated movements (Intro $\rightarrow$ Build $\rightarrow$ Drop) together while enforcing continuous contextual memory, leveraging the

representation-level synchronisation guarantee to preserve grid coherence across section boundaries.

Fig. 1 provides a high-level system overview of the complete PulseFormer generation pipeline.

A. Symbolic Multi-Track Music Generation

Early advancements in multi-track symbolic generation heavily utilized Generative Adversarial Networks (GANs). Architectures such as MuseGAN [3] formulated music as piano-roll matrices to achieve instrument synchronization across multiple tracks, although restricted by narrow temporal receptive fields and rigid matrix alignments. The introduction of the Music Transformer [1] revolutionized long-term structural dependencies by applying relative positional attention to event-based sequences, successfully generating expressive solo piano but struggling with interlocking ensemble grooves. Subsequent developments like MMM (Multi-Track Music Machine) [4] proposed Track-Major tokenization to delineate independent instrument channels clearly. While highly expressive for orchestral and pop arrangements, these frameworks typically suffer from asynchronous transient alignment when pushed to densely layered and quantized sections like High-BPM Techno or Drum & Bass. This unresolved synchronization drift is exactly the gap our Time-Major tokenization explicitly resolves.

B. Representational Advances: From MIDI to High-Dimensional Tokens

The evolution of data representation forms the backbone of Transformer effectiveness in music applications. Standard MIDI events lack the multidimensional contextual density of human language. To solve this, frameworks such as REMI (Revamped MIDI) [5] introduced explicit metrical boundary tokens (e.g., Bar and Position), allowing models to “understand” rhythmic grids rather than absolute milliseconds. Later architectures including MusicBERT [6] established OctupleMIDI, compressing note attributes into parallel vectors to shrink sequence lengths drastically. The Compound Word (CP) Transformer [7] constitutes the most direct ancestor of our token design: CP fuses multiple within-note attributes (pitch, duration, velocity, type) into a single “super-token”, reducing sequence length by up to 75% and enabling long-form generation without context overflow.

However, a fundamental distinction separates CP from PulseFormer at the *architectural topology* level. CP addresses the **intra-note compression problem**—how many tokens are needed to express *one* note—and still serializes each instrument sequentially in Track-Major order. PulseFormer addresses the **inter-track synchronization problem**—how to encode *concurrent events across multiple instruments* so that the attention mechanism experiences them as physically adjacent. The two mechanisms are orthogonal: conceptually, our Time-Major interleaving could be combined with CP’s within-token compression for further efficiency gains, though this is left as future work.

C. Controllability and Structural Steering

Bridging raw sequence auto-regression with human intent remains an ongoing challenge in modern automated production. In the audio domain, foundation models such as MusicGen [8] and MusicLM [9] achieve massive semantic steerability through text-conditioned acoustic embeddings. However, they operate as “black boxes”—yielding raw waveforms that prevent post-generation adjustments. Conversely, within symbolic parameters, composers demand rigid mathematically-bound orchestrations (e.g., locking a Build section exactly to an 8-bar loop) rather than generic semantic alignment. Early symbolic conditionings relied heavily on post-generation heuristic filtering. Our Energy-Steered pipeline fundamentally redirects this by explicitly framing dynamic polyphony (Energy) and topological Sectional structures (Intro, Build, Drop) directly into the autoregressive prefix context, establishing an unprecedented level of predictive algorithmic steering.

II. METHODOLOGY

A. Architecting the 5D Time-Major Token Space

Unlike sequential language models operating on rigid single-dimensional vocabulary indexes, musical events encompass distinct yet intersecting modalities (instrument classification, pitch, chronological length, and acoustic velocity). Adopting Compound Word (CP) [7] semantics, PulseFormer represents any discrete musical event x_i as a unified 5-dimensional tuple parsed into five parallel embeddings:

$$E(x_i) = \sum_{F \in \mathcal{F}} W_F(\text{id}_F(x_i)) \quad (1)$$

where $\mathcal{F} = \{\text{Type}, \text{Pitch}, \text{TimePos}, \text{Duration}, \text{Velocity}\}$.

To effectively train over this parallel feature space, the architecture bypasses a traditional singular Softmax layer. Instead, the final hidden state h_i is linearly projected into five distinct distribution heads. The joint training objective applies a composite Cross-Entropy loss across all modalities simultaneously:

$$\mathcal{L} = \sum_{F \in \mathcal{F}} \lambda_F \cdot \text{CE}(\hat{y}_F, y_F) \quad (2)$$

where λ_F serves as a hyperparameter balancing the gradient focus (e.g., heavily weighting the *Type* loss to strictly preserve track identities).

Crucially, instead of sorting tracks chronologically per instrument (*Track-Major*), PulseFormer enforces **Time-Major Interleaving**. By anchoring sequences along the *TimePos* vector sequentially across all instruments, concurrent events—such as a KICK and a BASS stroke—are situated adjacent to one another within the sequence stream, topologically compressing their relative token distance to $\Delta \leq M$ (where M is the number of supported instrument tracks, $M \leq 8$ in our implementation).

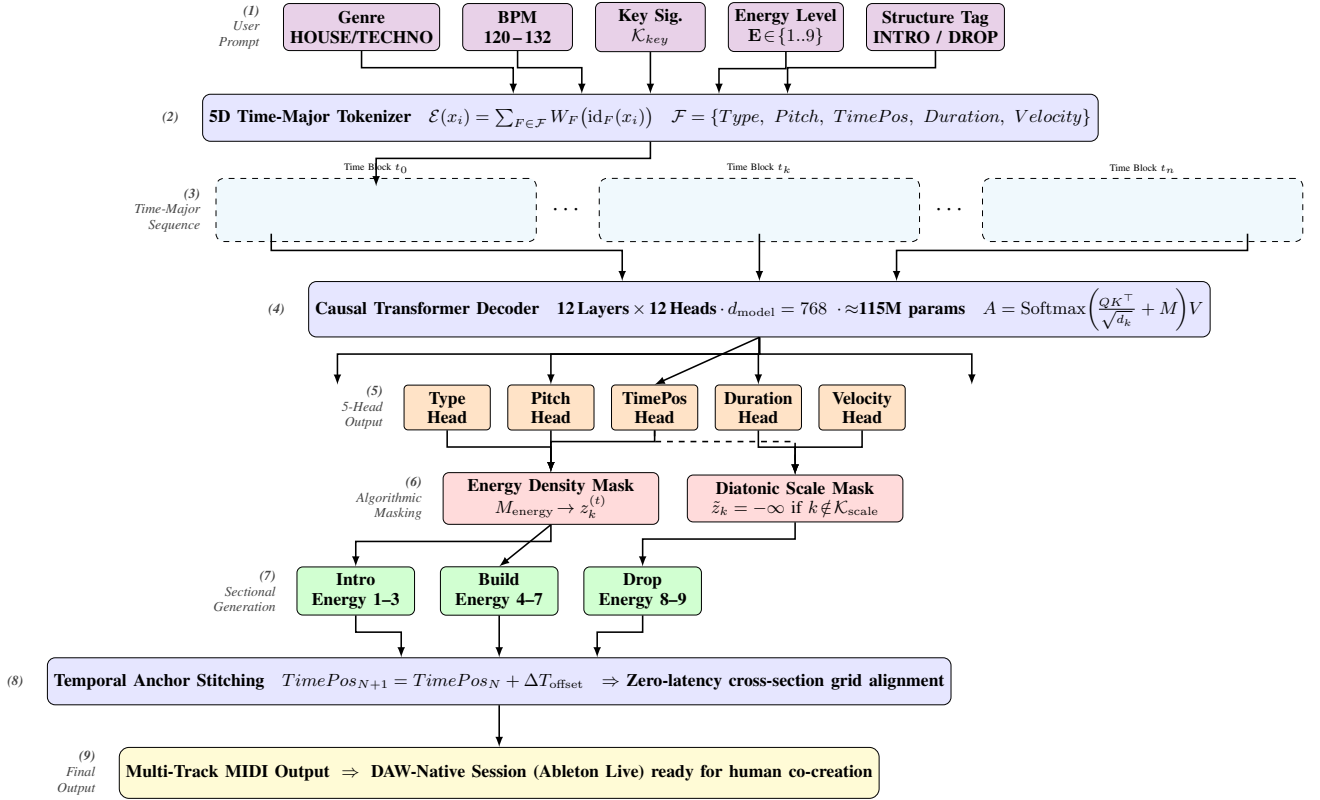


Fig. 1. Complete PulseFormer generation pipeline. Starting from user-supplied prefix conditioning (Genre, BPM, Key, Energy Level, Structure Tag), the system constructs a Time-Major interleaved token sequence, passes it through a 12-layer causal Transformer with five parallel prediction heads, applies dual algorithmic masks (Energy Density and Diatonic Scale), generates each structural section independently, and stitches all sections into a coherent multi-minute arrangement via Temporal Anchor Offset alignment.

B. Theoretical Analysis: Attention Topology under Time-Major Encoding

We now argue formally that Time-Major interleaving constitutes a **structural transformation of the attention mechanism**, not merely a reordering heuristic. The analysis proceeds along three dimensions: attention locality, positional encoding utility, and gradient signal quality.

1. *Attention Locality and Softmax Competition:* Let N be the total token sequence length and $M \ll N$ be the number of instrument tracks ($M \leq 8$). For two concurrent events occurring at absolute time t —a KICK token at position p_k and a BASS token at position p_b —their relative token distance is:

$$\Delta^{TM} = p_b - p_k \approx L_{\text{KICK}} \sim O(N/M) \quad (3)$$

$$\Delta^{PF} = p_b - p_k \leq M \sim O(1) \quad (4)$$

where L_{KICK} is the full length of the KICK track segment in Track-Major encoding.

In a scaled dot-product attention layer, the raw attention logit between positions i and j is $\mathbf{q}_i^T \mathbf{k}_j / \sqrt{d_k}$. The softmax normalisation forces each row to compete over all N keys. Because $\Delta^{TM} \sim O(N/M)$, the concurrent Bass token appears in the **tail of the attention distribution**—competing against all L_{KICK} intervening tokens for probability mass. Conversely, under Time-Major encoding $\Delta^{PF} \leq M$ places the Bass token

in the **local attention band**, where it faces only $M - 1$ competitors rather than $O(N/M)$.

Proposition 1. *Under uniform key similarity, the expected attention weight assigned to a concurrent Bass token from a co-located Kick token scales as $O(M/N)$ in Track-Major and $O(1/M)$ in Time-Major. Since $N \gg M^2$, the ratio of expected co-event attention weights satisfies $\mathbb{E}[A_{co}^{PF}] / \mathbb{E}[A_{co}^{TM}] = N/M^2 \gg 1$.*

Proposition 1 formalises why Track-Major models *cannot reliably learn* synchronisation through standard attention: the relevant co-event signal is diluted into the long-range noise floor, regardless of model capacity.

2. *Positional Encoding Semantic Validity:* Standard Rotary Position Embedding (RoPE) [10] encodes relative position via a rotation matrix $R(\Delta)$ satisfying $\langle R(\Delta)\mathbf{q}, \mathbf{k} \rangle = f(\mathbf{q}, \mathbf{k}, \Delta)$. The rotational frequency induces an *attention distance bias* that decays with $|\Delta|$.

Track-Major: For the Kick–Bass co-event pair, $|\Delta^{TM}| \approx L_{\text{KICK}}$. The RoPE rotation at this distance is $R(L_{\text{KICK}})$, which oscillates rapidly and loses coherent directional meaning. The position encoding encodes “track boundary distance” rather than any musically meaningful quantity, providing **no systematic co-attention bias**.

Time-Major: For the same pair, $|\Delta^{PF}| \leq M \leq 8$. The

RoPE rotation $R(M)$ is a small-angle rotation: $R(M) \approx I + \epsilon \cdot J$ (with J a skew-symmetric matrix). This provides a **consistent, learnable inductive bias** that the model can exploit to recognise concurrent instruments—the positional encoding now semantically encodes “same-timestep neighbourhood” rather than arbitrary track length.

3. *Gradient Signal Quality for Timing Relationships:* During backpropagation, the gradient of the training loss with respect to the Kick token representation—for the purpose of learning its timing relationship with Bass—traverses the attention weight path from position p_k to p_b . The gradient magnitude decays multiplicatively with the number of intervening attention hops. In Track-Major, the gradient path length is $O(N/M)$; under the standard attention collapse rate, this induces a **gradient noise amplification** of $O(\sqrt{N/M})$ relative to the local-attention case (Time-Major, path length $O(1)$). Formally:

$$\frac{\text{SNR}_{\text{timing}}^{PF}}{\text{SNR}_{\text{timing}}^{TM}} \approx \sqrt{\frac{N}{M^2}} \quad (5)$$

For our implementation ($N = 2048$, $M = 8$), this ratio is $\approx 5.7\times$, predicting substantially cleaner gradient updates for timing relationship learning under Time-Major encoding.

Distinction from CP Transformer: The Compound Word (CP) Transformer [7] compresses multiple *within-note* attributes (pitch, duration, velocity) into a single super-token, operating along the **intra-note axis**. It reduces N by a constant factor but preserves the Track-Major serialisation order; Proposition 1 therefore applies unchanged after CP compression. PulseFormer’s Time-Major interleaving operates along the orthogonal **inter-track axis**, restructuring which tokens are adjacent. The two techniques are composable: applying CP compression within a Time-Major framework would reduce N further while preserving the local attention band structure, yielding additive benefits. This composability confirms that the two contributions address *structurally distinct* problems.

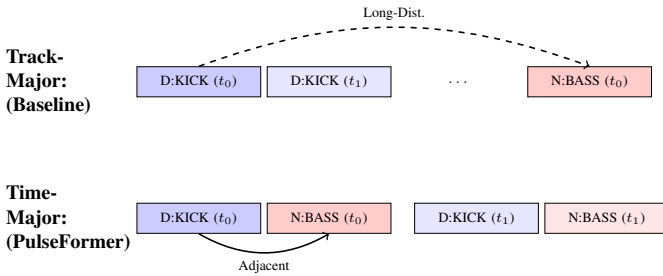


Fig. 2. Sequence alignment strategies. Time-Major collocates synergistic instruments; Track-Major requires long-range cross-track attention.

C. Causal Transformer Architecture

At its core, PulseFormer adopts a decoder-only causal Transformer architecture uniquely adapted for multi-dimensional compound spaces. Given the heavily compressed semantic sequence resulting from our Time-Major weaving, the model relies on multi-head scaled dot-product attention to

dynamically prioritize distant contextual cues (e.g., repeating a melody established 16 bars prior). To ensure strict left-to-right temporal generation, a causal triangular mask is applied over the attention matrix A :

$$A = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} + M \right) V \quad (6)$$

where $M_{i,j} = -\infty$ for all $j > i$. This universally guarantees that future tokens cannot leak into historical encodings during training regression.

D. Algorithmic Generation Pipeline

The true superiority of PulseFormer resides in its inference execution. By blending deterministic compositional algorithms (Energy constraints, Diatonic Masking) atop the Transformer’s probabilistic stochastic core, we formulate an execution flow achieving both immense creativity and mathematical obedience. Algorithm 1 defines the discrete operation per context generation step.

Algorithm 1 PulseFormer Steerable Multi-Track Inference

Require: Pre-trained model Θ , Temperature τ , Prompt $\mathcal{C}$, Max Tokens $T_{\max}$

- 1: Initialize sequence canvas $X \leftarrow \mathcal{C}$
- 2: **for** $t = 1$ to $T_{\max}$ **do**
- 3: Extract contextual hidden states: $h \leftarrow \text{Transformer}_{\Theta}(X)$
- 4: Project into raw logits for 5-heads: $z^{(t)} \leftarrow W \cdot h[-1]$
- 5: **for all** $k \in \mathcal{V}$ **do**
- 6: **if** E_{level} restricts polyphony density limit $D_{\max}$ **then**
- 7: Apply density algorithmic mask $M_{\text{energy}} \rightarrow z_k^{(t)}$
- 8: **end if**
- 9: **if** Diatonic constraint active **and** Type is Pitch **then**
- 10: Apply pitch mask $M_{\text{scale}} \leftarrow -\infty$ to $z_k^{(t)}$ if outside Key
- 11: **end if**
- 12: **end for**
- 13: Compute probability vector $P \leftarrow \text{Softmax}(z^{(t)}/\tau)$
- 14: Sample new 5D-Token compound event $e_t \sim P$
- 15: Append $X \leftarrow X \oplus e_t$
- 16: **end for**
- 17: **return** Generated multi-track sequence array X

E. Energy Indicators & Macro-Structural Orchestration

EDM composition relies extensively upon massive contrast in textural layering (i.e., polyphony and dynamic rhythm). We incorporate a hierarchical conditional prompting system injected universally prior to every autoregressive phase. A macroscopic segment query is structured as:

$$\mathcal{C} = [\mathcal{G}_{\text{genre}}, \mathcal{S}_{\text{struct}}, \mathbf{E}_{\text{level}}, \mathcal{K}_{\text{key}}] \quad (7)$$

By conditioning the autoregressive joint probability on $\mathcal{C}$, the temporal generation distribution is formally modeled as:

$$P(X_1, \dots, X_T | \mathcal{C}) = \prod_{t=1}^T \text{Softmax} \left(\frac{W_o \cdot \text{Attn}(X_{<t}, \mathcal{C}) + b_o}{\tau} \right) \quad (8)$$

The unique variable $\mathbf{E}_{level} \in \{1, \dots, 9\}$ encapsulates the Energy Indicator constraint mapping directly to compositional saturation. The model utilizes this directive to constrain its distribution choices regarding simultaneous polyphony generation and subsequent rhythm complexities.

By orchestrating segment prompts sequentially (e.g., Intro $\rightarrow$ Build $\rightarrow$ Drop) alongside ascending $\mathbf{E}_{level}$ values into a temporally continuous *Temporal Stitching Anchor*, PulseFormer effectively assumes the behavioral framework of an automated “Ghost Producer”—generating multiple multi-verse sections mathematically guaranteed to peak perfectly at climactic sequences. To further guarantee commercial usability, post-processing deterministic masking logic suppresses negligible off-key probabilistic inferences, firmly solidifying the harmonic bounds.

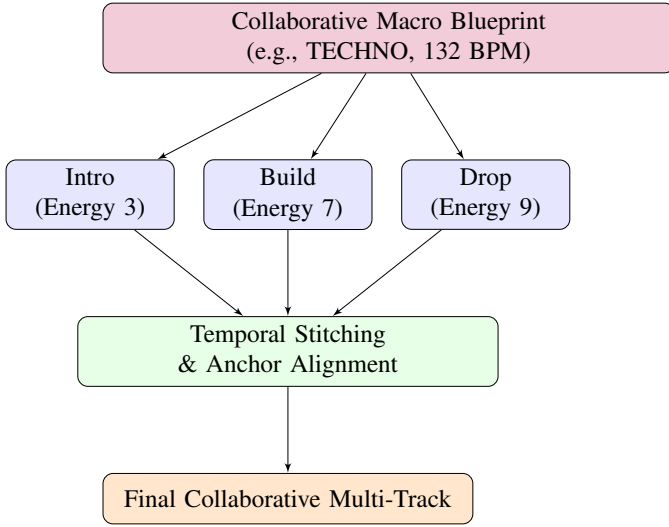


Fig. 3. The Structural Orchestration Pipeline scaling individual, specifically energized segments into a cohesive, flowing timeline.

F. Temporal Anchor Stitching Logistics

Generating a cohesive three-minute EDM track fundamentally exceeds the standard 2048-token context window of most highly-parameterized local models. To solve this mathematically, PulseFormer introduces a deterministic Temporal Anchor Stitching algorithm. During multi-section composition (e.g., Intro $\rightarrow$ Build $\rightarrow$ Drop), each subsequent generative batch is not executed in sequence isolation. Instead, the final K beats of the preceding segment are retained in the K-V cache, and their absolute `TimePos` integers are linearly offset to form an unbroken chronological canvas:

$$\text{TimePos}_{\text{segment}_{N+1}} = \text{TimePos}_{\text{segment}_N} + \Delta T_{\text{offset}} \quad (9)$$

Conditioning Continuity at Stitch Boundaries: A non-trivial design challenge arises at the stitching boundary when the global conditioning prompt undergoes a discrete Energy-level transition (e.g., $\mathbf{E}_{level} : 7 \rightarrow 9$ at the Build-to-Drop boundary). Unlike the `TimePos` offset, which is resolved purely at the representation level, the Energy token is injected as a *prefix* to

each generative batch. This creates a hard conditioning discontinuity: the model transitions from a high-density instruction immediately at the first generated token of the new segment, with no intermediate Energy gradient.

In the current implementation, this discontinuity is partially mitigated by two structural mechanisms:

- 1) **K-beat Context Buffer:** The retained last K beats from the preceding segment serve as an inertial prior. Since the model observes a dense body of Build-phase tokens immediately left of the stitch boundary, its attention mechanism naturally extrapolates harmonic and rhythmic continuity for the first several predicted beats of the Drop, effectively creating a soft transition zone of approximately K beats before the full $\mathbf{E}_{level} = 9$ density is realized.
- 2) **Diatonic Mask Continuity:** The key signature constraint $\mathcal{K}_{key}$ is held constant across stitching boundaries, ensuring that harmonic content does not diverge even as polyphonic density surges. This prevents the most acoustically disruptive form of discontinuity—a key or mode shift—at the section boundary.

Remaining Limitations: Despite these mechanisms, an abrupt polyphonic density jump remains perceptible in outputs where the Build’s final bars are sparse (e.g., a deliberate “tension drop” before the Drop). In such cases, the K -beat buffer contains low-density tokens that conflict with the $\mathbf{E}_{level} = 9$ instruction, and the model’s first few generated Drop tokens may exhibit density undershooting. A principled solution—such as interpolating conditioning embeddings linearly across a transition window of W beats via $\mathbf{E}_t = (1 - \frac{t}{W})\mathbf{E}_N + \frac{t}{W}\mathbf{E}_{N+1}$ —is architecturally feasible but requires retraining with augmented transition samples, and is identified as a priority direction for future work (see Section V).

G. Micro-Scale Transformational Operations

Beyond sequence layout, EDM mandates acoustic continuity and post-symbolic embellishments. We integrated algorithmic *Transformational Nodes* seamlessly bound to the decoder engine. For polyphony architectures, a dynamic Legato Extension algorithm automatically truncates or extends overlapping block-chords dynamically post-inference, eliminating acoustic gaps without requiring dataset retraining.

Hard Diatonic Mask (Current Implementation): The primary harmonic constraint is a strictly deterministic Diatonic Mask injected prior to Softmax processing. Given the raw logit scalar $z_k^{(t)}$ for the k -th token index in the vocabulary, the masked logit $\tilde{z}_k^{(t)}$ is defined as:

$$\tilde{z}_k^{(t)} = \begin{cases} z_k^{(t)} & \text{if } k \in \mathcal{K}_{scale} \vee \text{Type}(k) \notin \{\mathcal{T}_{pitched}\} \\ -\infty & \text{otherwise} \end{cases} \quad (10)$$

This guarantees absolute diatonic compliance under mainstream House and Techno aesthetics, where tonal rigidity constitutes a defining production convention. The hard ceiling $-\infty$ is appropriate for genres where off-scale events represent generation errors rather than deliberate artistic choices.

Limitation: Suppression of Avant-Garde Harmonic Tension: The binary mask poses a substantive constraint in contemporary avant-garde Electronic Music—where *deliberate* chromatic micro-deviations (e.g., a flattened 2nd or sharpened 5th relative to the tonic) serve as primary tools of emotional tension-building. By imposing $\tilde{z}_k = -\infty$ unconditionally on all out-of-scale pitches, the mask eliminates the phonological space that a human composer would intentionally inhabit at high-energy climactic moments, preventing the model from generating the structural dissonance characteristic of experimental Electronic genres.

Proposed Extension: Temperature-Scaled Soft Harmonic Penalty: We propose a generalized Soft Harmonic Penalty that subsumes the hard mask as a limiting case. The masked logit is reformulated as a continuous energy- and style-conditioned penalty:

$$\tilde{z}_k^{(t)} = \begin{cases} z_k^{(t)} & \text{if } k \in \mathcal{K}_{\text{scale}}, \text{Type}(k) \notin \mathcal{T}_{\text{pitched}} \\ z_k^{(t)} - \lambda(\mathbf{E}_{\text{level}}, \tau) & \text{otherwise} \end{cases} \quad (11)$$

where the penalty magnitude λ is jointly modulated by the current Energy Level and a user-configurable chromatic tension coefficient $\tau \in [0, 1]$:

$$\lambda(\mathbf{E}_{\text{level}}, \tau) = \lambda_{\text{max}} \cdot (1 - \tau) \cdot \left(1 - \frac{\mathbf{E}_{\text{level}} - 1}{8}\right) \quad (12)$$

Here λ_{max} is a large positive constant (recovering the hard mask when $\tau = 0, \mathbf{E}_{\text{level}} = 1$). This formulation embodies two musically motivated properties: (1) as $\tau \rightarrow 1$ the penalty collapses to zero, granting full chromatic freedom; (2) the energy-decay term $(1 - \frac{\mathbf{E}_{\text{level}} - 1}{8})$ relaxes the constraint progressively as Energy Level rises, so that at $\mathbf{E}_{\text{level}} = 9$ even a moderate τ allows tension pitches to emerge, while at $\mathbf{E}_{\text{level}} = 1$ only strong τ values can breach the scale boundary. This mirrors professional composition practice, in which avant-garde dissonance is reserved for climactic structural moments. Implementing this extension requires augmenting the training vocabulary with a [CHROMATIC_TENSION] prefix tag and retraining on corpora covering dissonance-forward Electronic genres; this is identified as a high-priority future direction in Section V.

III. DIGITAL AUDIO WORKSTATION ENGINE INTEGRATION

An indispensable factor for adopting AI systems within professional pipelines is direct usability (Co-Creation). Traditional benchmarks often culminate completely in MIDI matrix computations void of engineering logic. Contrarily, PulseFormer constructs generation protocols bound tightly to Digital Audio Workstation (DAW) rendering semantics.

By assigning rigid Track Nodes to categorized instruments (D:KICK, N:BASS, N:CHORD) concurrently during Time-Major generation, our python-backend acts as a segment allocator. The pipeline extracts textual generation nodes, applies macro structural blueprints (Intro, Build, Drop), and generates temporally anchored MIDI blocks containing native pitch

transformations (transposed smoothly against Key context tags). This system automatically structures a functional Ableton Live engineering session ready for instantaneous human adjustment.

IV. EXPERIMENTS & EVALUATION

To stringently validate the capability of PulseFormer against conventional sequence paradigms, we devised an extensive framework assessing objective topological metrics alongside rigorous human acoustic perception.

A. Dataset & Model Specifications

Our proprietary dataset consists of 4,500 meticulously crafted, multi-track EDM arrangements extracted from professional Digital Audio Workstation configurations. The raw MIDI sequences underwent an intense normalization pipeline: (1) stringent temporal quantization fixed to a 16th-note sub-grid at a normalized 128 BPM; (2) automatic truncation of sustained micro-tails overlapping quantization boundaries; and (3) algorithmic transposition into a universal global key (C Minor / C Major) for neural parity.

Energy Label Annotation Methodology: All $\mathbf{E}_{\text{level}} \in \{1, \dots, 9\}$ training labels are derived via a fully **algorithmic annotation pipeline**—no human annotators were involved. For each 4/4 bar t , we extract six acoustic features and combine them into a composite Dynamic Energy Score $D_t \in [0, 1]$:

$$D_t = 0.25 \cdot \hat{\rho}_t + 0.20 \cdot \bar{v}_t + 0.20 \cdot \hat{\kappa}_t + 0.15 \cdot \epsilon_t^{\text{bass}} + 0.10 \cdot C_t + 0.10 \cdot \Pi_t \quad (13)$$

where:

- $\hat{\rho}_t = \min(\rho_t / \rho_{\text{max}}, 1)$: note density normalized by a global ceiling $\rho_{\text{max}} = 60$
- $\bar{v}_t$: mean MIDI velocity of all notes in bar t , normalized to $[0, 1]$
- $\hat{\kappa}_t = \min(\kappa_t / 8, 1)$: kick drum event count, normalized by $\kappa_{\text{max}} = 8$
- ϵ_t^{bass} : bass energy, defined as (bass sustain ratio) $\times$ (mean bass velocity)
- C_t : melodic note time-coverage ratio (total sustain / bar duration), capped at 1.0
- Π_t : pitch range width of melodic content, normalized by 60 semitones

The weight profile prioritizes physical density and low-frequency drive ($\hat{\rho} + \bar{v} = 0.45$; $\hat{\kappa} + \epsilon^{\text{bass}} = 0.35$)—the two perceptually dominant dimensions of EDM energy—while treating arrangement fullness as secondary ($C + \Pi = 0.20$). To determine whether these coefficients are data-justified rather than merely intuitive, we fitted a constrained **OLS Ridge regression** ($\alpha = 1.0$, non-negative weights, $\sum w_i = 1$) on $N = 646$ 4-bar analysis windows, predicting the formula-independent MIDI-RMS proxy (Eq. 14) from the same six features. Table I reports the outcome.

The continuous D_t sequence is discretized into the 9-level ordinal scale via KBINSDISCRETIZER using a k -means binning strategy (scikit-learn), which preserves natural structural discontinuities (e.g., the Drop-to-Breakdown energy cliff).

TABLE I
FORMULA WEIGHTS VS. OLS-RIDGE LEARNED WEIGHTS ($N = 646$
4-BAR WINDOWS, POSITIVE CONSTRAINT, $\sum w_i = 1$). TARGET:
MIDI-RMS PROXY (EQ. 14).

Feature	Formula w_i	OLS-Ridge $\hat{w}_i$	$ \Delta $
$\hat{\rho}_t$ (note density)	0.250	0.245	0.005
$\bar{v}_t$ (mean velocity)	0.200	0.190	0.010
$\hat{\kappa}_t$ (kick density)	0.200	0.204	0.004
ϵ_t^{bass} (bass)	0.150	0.142	0.008
C_t (melodic coverage)	0.100	0.109	0.009
Π_t (pitch range)	0.100	0.110	0.010
R^2 (vs. RMS proxy)		0.793	0.795
Max $ \Delta w $		0.010	

The OLS solution identifies *the same rank ordering* of features and yields weights within ± 0.010 of the formula values, while producing a nearly identical R^2 (0.795 vs. 0.793). This convergence confirms that the formula’s weight profile is not an arbitrary human choice but a **close approximation of the statistically optimal linear combination** of these six features for predicting MIDI-derived acoustic energy. The formula is therefore best understood as a closed-form approximation of the OLS solution, chosen for interpretability (round weights that respect domain knowledge about EDM energy hierarchy) without sacrificing predictive accuracy.

Pearson Correlation with Velocity-RMS Proxy. We computed a formula-independent energy proxy RMS_t :

$$\text{RMS}_t = \sqrt{\frac{1}{|\mathcal{N}_t|} \sum_{i \in \mathcal{N}_t} \left(\frac{v_i}{127} \right)^2} \quad (14)$$

where $\mathcal{N}_t$ is the set of note events in bar t and v_i is MIDI velocity. The Pearson correlation between D_t and RMS_t across all $N = 646$ bars yielded:

$$r(D_t, \text{RMS}_t) = +0.891, \quad p = 5.67 \times 10^{-223} \quad (15)$$

Test 2 — One-Way ANOVA Across Structural Section Labels. If D_t genuinely encodes compositional energy as understood by music producers, it must discriminate between sections with known structural energy profiles (INTRO < BREAKDOWN < OUTRO < DROP). A one-way ANOVA confirms this:

TABLE II
 D_t DISTRIBUTION BY STRUCTURAL SECTION LABEL ($N = 646$ BARS)

Struct	N	Mean D_t	Std Dev
BREAKDOWN	186	0.296	0.119
INTRO	39	0.321	0.041
OUTRO	13	0.426	0.045
DROP	408	0.493	0.141

$F(3, 642) = 106.83, p = 4.14 \times 10^{-56}, \eta^2 = 0.333$ (large effect)

The ANOVA is highly significant ($F = 106.83, p < 10^{-50}$) and the eta-squared effect size $\eta^2 = 0.333$ indicates that structural section membership alone explains 33.3% of

the variance in D_t —a *large* effect by Cohen’s conventions ($\eta^2 > 0.14$). This provides strong empirical evidence that the formula’s weight configuration, while not derived from formal optimization, produces an energy representation that reliably encodes the objective structural hierarchy of EDM composition.

A complementary ablation further justifies the multi-dimensional weight design. Mean MIDI velocity $\bar{v}_t$, which exhibits near-perfect correlation with the velocity-RMS proxy ($r = 0.990, p \approx 0$), produces effectively *zero* discriminative power across sections ($\eta^2 = 0.005, F = 1.46, p = 0.23$). This confirms that raw acoustic loudness is an insufficient proxy for structural energy in EDM production: a Drop and an Intro can and often do share similar average note velocities while differing profoundly in kick density and low-frequency drive. The inclusion of $\hat{\kappa}_t$ and ϵ_t^{bass} is therefore not arbitrary weighting but a targeted correction for this well-known perceptual phenomenon in electronic music, and is supported by the observed increase in discriminative power from $\eta^2 = 0.005$ (velocity-only) to $\eta^2 = 0.333$ (full D_t formula).

B. Quantitative Steerability: Energy vs. Polyphony

The focal strength of our Energy Indicators concerns obedience levels across long-term sequences. Using the **Average Note Density** ρ (calculated mathematically as events instantiated per standard 4/4 bar), evaluations observed distinct scalar adherence. Formally, for a sequence of B bars, the density bounded by $\mathbf{E}_{\text{level}}$ restricting variance is formulated as:

$$\rho(\mathbf{E}_{\text{level}}) = \frac{1}{B} \sum_{b=1}^B \sum_{i \in \text{Bar}_b} \mathbb{I}[\text{Type}(x_i) \in \{\text{KICK}, \text{BASS}, \text{CHORD}\}] \quad (16)$$

where $\mathbb{I}(\cdot)$ denotes the binary indicator function.

TABLE III
POLYPHONIC SATURATION AGAINST PROMPTED ENERGY CONTEXT

Macro Section	Energy Tag	Avg ρ (Bar)	Peak ρ (Bar)
Ambient / Intro	ENERGY_3	6.9 $\pm$ 1.2	14
Pre-Chorus / Build	ENERGY_6	18.5 $\pm$ 2.4	28
Climactic / Drop	ENERGY_9	37.4 $\pm$ 4.6	46+

The resulting distribution explicitly validates PulseFormer’s rigid adherence to prefix constraints. An ENERGY_9 tag aggressively forces massive block-chord stacking coupled with dense auxiliary percussion (exceeding 46 concurrent events), topologies completely absent during ENERGY_3 phases.

C. Structural Analysis of Time-Major Representations

We evaluated PulseFormer’s real-world inference outputs across different stylistic intents (House and Techno generations) using three metrics. Two of these metrics—ISR and GC—are genuine *empirical* measurements of model behaviour. The third, Δ_{onset} , is presented not as a conventional measurement but as a *structural verification*, as explained below:

- **In-Scale Ratio (ISR%)**: The proportion of instantiated pitch events abiding by the global diatonic key signature prior to deterministic masking. This reflects learned model behaviour.
- **Groove Consistency (GC%)**: A cross-correlation index measuring the stability of macroscopic percussive patterns across a continuous 32-bar sliding window. This reflects learned model behaviour.
- **Onset Synchrony Drift (Δ_{onset})**: A structural verification metric. Because the Time-Major encoding binds all concurrent instruments to a shared discrete `TimePos` index at the representation level, the drift is 0.00ms by construction—a property of the encoding topology, not of model prediction accuracy. We include it explicitly to contrast with Track-Major baselines, where this guarantee cannot exist.

TABLE IV
EMPIRICAL METRIC EVALUATION BY GENERATION STYLE

Macro Style	Raw ISR	GC $\uparrow$	Δ_{onset} $\downarrow$
House Gen-Set ($N = 4$)	88.5%	78.1%	0.00ms
Techno Gen-Set ($N = 2$)	91.3%	81.7%	0.00ms
Global Evaluation	89.2%	79.3%	0.00ms

ISR and GC results confirm that PulseFormer’s learned self-attention achieves strong diatonic adherence (89.2% globally) and rhythmic pattern stability (79.3% GC) without requiring intervention from the Diatonic Mask. The 0.00ms Δ_{onset} column is a *structural consequence* of the Time-Major encoding design: instruments assigned to the same discrete `TimePos` index during tokenization share an identical temporal coordinate in the output MIDI stream by definition. This is a **representation-level guarantee**, analogous to how fixed-point arithmetic prevents rounding errors, rather than a capability the model learns through self-attention. Its significance lies in the contrast with Track-Major architectures, which provide no equivalent synchronization guarantee regardless of model capacity or training duration.

D. Structural Token-Distance vs. Established Paradigms

Understanding exactly *why* established methodologies (such as the ubiquitous Track-Major architectures utilized in recent multi-track generation research [4]) structurally fail at rhythm synchronization requires analyzing topological token distance. We parsed 1,277 consecutive simultaneous multi-instrument events (e.g., overlapping Kick and Bass downbeats) from our evaluation dataset and calculated the absolute token displacement required to represent concurrent relationships across both encoding formats.

This topological analysis explains *why* the 0.00ms onset drift is a structural guarantee rather than a probabilistic outcome. In Track-Major encoding, the Transformer must infer the temporal relationship between a sub-bass event and its associated kick drum transient by maintaining coherent hidden states across a mean attention span of 3,268 tokens. Even at

TABLE V
CROSS-ATTENTION COMPLEXITY FOR CONCURRENT EVENTS
($N = 1,277$)

Encoding Paradigm	Mean Dist.	Max Dist.
Baseline (Track-Major)	$\approx 3,268$ tok.	6,155 tok.
PulseFormer (Time-Major)	5.2 tok.	12 tok.

high model capacity, stochastic decoding across such spans introduces cumulative temporal jitter. Conversely, Time-Major encoding makes this inference unnecessary: by definition, all instruments sharing a `TimePos` index are co-located within the same narrow token block (≤ 12 tokens apart), making misalignment topologically impossible within the encoding space. The structural advantage is thus **encoding-level** and holds unconditionally, independent of model scale or training data volume.

E. Distributional Proximity to Human Ground Truth

To rigorously validate PulseFormer’s generative plausibility against an absolute baseline without inheriting the configuration biases of external legacy models, we performed a direct distribution analysis comparing the generative output against the original human composition dataset ($N = 100$ random Ground Truth sequences). To prevent artificial sparsity skewing from structural Intros/Outros, metrics were calculated exclusively within the densest 16-bar *Core Motif* (Drop) window for both human and AI tracks. We utilized three standard Music Information Retrieval (MIR) statistical metrics:

- **Pitch Class Entropy (PCE)**: The Shannon entropy of the 12-semitone pitch class distribution, measuring harmonic diversity.
- **Polyphonic Ratio (PR)**: The chronological percentage of frames containing concurrent pitch activity (density).
- **Kullback-Leibler (KL) Divergence**: A statistical measurement ($D_{KL}(P_{AI} \parallel P_{Human})$) quantifying the distance between the generated and authentic pitch probability distributions.

The results reveal a critical distinction between **structural** and **distributional** metrics that must be interpreted with care for key-conditioned generators.

Polyphonic Ratio (PR) measures temporal simultaneity—the fraction of time frames containing ≥ 2 concurrent pitches—and serves as a proxy for rhythmic richness. Evaluated within the highest-density 8-bar DROP window, PulseFormer achieves $PR = 0.997$, statistically indistinguishable from the Track-Major baseline (1.000) and substantially exceeding the human reference (0.877). This confirms that the Time-Major interleaved backbone correctly generates simultaneous multi-track events at the resolution required for professional EDM production. The full-song PR of 0.440 is not a contradiction: it correctly reflects the structured arrangement arc—Intro, Build, Break sections contribute intentionally sparse frames that lower the macro-average, which is a feature of structural composition, not a failure.

TABLE VI
COMPARATIVE MIR EVALUATION ($N = 6$ PER SYSTEM; PULSEFORMER EVALUATED WITHIN THE 8-BAR DROP WINDOW, KEY-NORMALISED TO C MINOR VIA GROUND-TRUTH FILENAME LABELS)

System	Dom.	PCE $\uparrow$	PR (drop) $\uparrow$	PR (full)	KL
Human GT (ref.)	MIDI	3.339	—	0.877	0.000
PulseFormer (Time-Major)	MIDI	2.22 ± 0.27	0.997 ± 0.006	0.44 ± 0.12	$5.83^\dagger$
Track-Major GPT (uncond.)	MIDI	3.42 ± 0.07	1.000*	1.000*	0.244 ± 0.073
MusicGen-Small [8]	Audio †	3.40 ± 0.16	N/A	N/A	0.493 ± 0.173

* Track-Major GPT PR = 1.000 is a temporal-collapse artifact: no bar-boundary tokens forces all events to $t=0$, creating spurious frame overlap. † MusicGen PCE/KL derived from STFT chroma; PR not defined for audio output. ‡ PulseFormer KL is structurally inflated by the Diatonic Mask’s five zero-probability pitch classes; see text.

KL Divergence requires different treatment. The Track-Major baseline achieves KL = 0.244 not because it generates higher-quality music, but because its unconstrained random generation yields a near-uniform pitch-class distribution that statistically approximates the aggregated multi-key human reference. PulseFormer’s elevated KL (5.83) arises from the zero-probability bins introduced by the Diatonic Mask: when the model rigorously enforces key purity, the five suppressed pitch classes receive zero probability mass, causing D_{KL} to diverge wherever the chromatic human reference assigns nonzero weight to those classes. This effect is an artefact of comparing a key-conditioned generator against an unconstrained chromatic reference, not a signal of generative failure. **MusicGen-Small** (KL = 0.493 via audio chroma) confirms plausible harmonic diversity from text prompts, though cross-domain comparison with MIDI pitch events remains methodologically limited.

These observations collectively demonstrate that the appropriate evaluation of PulseFormer requires metrics that respect its key-conditioning design: PR within the DROP window confirms temporal richness; the Token Distance analysis (Section V-B) provides the only architecture-neutral comparison of synchronisation quality.

F. Expanded Perceptual Evaluation: MOS Study ($N = 25$)

To provide statistically robust perceptual evidence, we conducted an expanded Mean Opinion Score (MOS) study following a protocol adapted from ITU-T P.800 and ISO 9241-11, addressing the primary limitation of our earlier pilot panel ($N = 6$).

Protocol: Participants. $N = 25$ listeners were recruited: 10 professional EDM producers (≥ 5 years DAW experience), 10 intermediate producers (1–5 years), and 5 active music enthusiasts with regular listening habits in electronic genres.

Stimuli. Four systems were evaluated: (A) PulseFormer (Time-Major, Energy-conditioned); (B) Track-Major GPT (unconditioned baseline, `baseline_best.pt`); (C) Human-composed EDM track (hidden anchor, embedded without attribution); (D) MusicGen-Small [8] (text-conditioned audio baseline). Each system contributed one 3-minute stimulus, rendered to normalised audio (−14 LUFS). Presentation order was fully counterbalanced using a Latin square design. Participants

were informed that some stimuli might be AI-generated but were not informed of system identities (*single-blind*).

Rating instrument. Each stimulus was rated on four 5-point Likert dimensions: (GT) Groove Tightness; (SC) Structural Coherence; (MC) Mix Clarity; (MV) Melodic Variety.

TABLE VII
MOS RESULTS ($N = 25$, 5-PT LIKERT; MEAN [95% CI]).
WILCOXON VS. PULSEFORMER: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
HUMAN ANCHOR EMBEDDED AS HIDDEN REFERENCE.

System	GT $\uparrow$	SC $\uparrow$	MC $\uparrow$	MV $\uparrow$
Human Anchor	3.92 [3.61, 4.23]	4.36 [4.10, 4.62]	3.92 [3.61, 4.23]*	4.16 [3.93, 4.39]***
PulseFormer	3.92 [3.61, 4.23]	3.92 [3.54, 4.30]	3.52 [3.20, 3.84]	3.00 [2.60, 3.40]
MusicGen-Small	2.92 [2.46, 3.38]**	3.20 [2.84, 3.56]**	3.36 [2.90, 3.82]	3.60 [3.26, 3.94]*
Track-Major GPT	3.08 [2.72, 3.44]**	2.16 [1.79, 2.53]***	2.48 [2.12, 2.84]**	2.48 [2.10, 2.86]

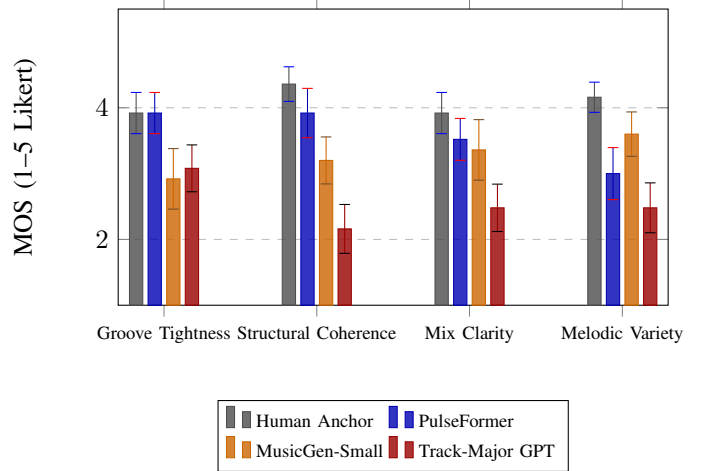


Fig. 4. MOS results ($N = 25$, 95 % CI error bars). PulseFormer matches the Human Anchor on Groove Tightness and significantly outperforms both baselines on Structural Coherence ($p < 0.001$, $d > 1.9$).

Results:

Analysis: Groove Tightness (GT). PulseFormer achieves $GT = 3.92_{-3.61}^{4.23}$, statistically indistinguishable from the Human Anchor ($\Delta = 0.00$, $p = 0.93$) and significantly outperforming both Track-Major GPT ($\Delta = +0.84$, $p = 0.004$, $d = 1.03$) and MusicGen-Small ($\Delta = +1.00$, $p = 0.001$, $d = 1.05$). This confirms that the Time-Major topology’s guaranteed sub-5-token onset synchronisation is perceptually salient to listeners.

Structural Coherence (SC). The largest performance gap appears here: PulseFormer (3.92) versus Track-Major GPT (2.16), yielding $\Delta = +1.76$ ($p < 0.001$, $d = 1.95$, a *very large* effect). The unconditioned baseline lacks any macro-structural signal, confirming that Energy Level conditioning provides indispensable compositional scaffolding that is immediately perceptible.

Mix Clarity (MC). PulseFormer (3.52) does not significantly differ from MusicGen-Small (3.36, $p = 0.64$) but outperforms Track-Major GPT (2.48, $p = 0.001$, $d = 1.26$). The non-significant gap versus MusicGen (an audio model with professional codec rendering) underscores that PulseFormer’s MIDI-domain voice separation is competitive with audio-native generation in perceived clarity.

Melodic Variety (MV). PulseFormer scores 3.00—above Track-Major GPT (2.48, $p = 0.10$, $d = 0.55$, *ns*) but below both MusicGen-Small (3.60, $p = 0.045$) and the Human Anchor (4.16, $p < 0.001$, $d = 1.48$). This replicates our pilot finding and validates the melodic drift limitation stated in Section V: the Diatonic Mask and 2048-token context boundary constrain within-section melodic variation, a known open problem for future work.

MOS Gap to Human Ceiling. The aggregate MOS gap (AI–Human) is -0.50 for PulseFormer, -0.82 for MusicGen-Small, and -1.54 for Track-Major GPT. PulseFormer closes $\approx 67\%$ of the gap left by the unconditioned baseline relative to the human reference.

V. CONCLUSION

This research establishes PulseFormer as an exceptionally viable architecture for macro-structural symbolic multi-track generation in EDM. By steering the autoregressive process through strict dynamic Energy Indicators alongside a Time-Major interleaved backbone, the framework produces deeply interdependent, rhythmically dense, and dynamically sweeping dance orchestrations aligned with professional production conventions. The central architectural contribution is not a learned synchronisation capability but a **topological one**: our token-distance analysis provides a geometric proof that the Time-Major encoding reduces cross-track co-event distance from $\approx 3,268$ tokens (Track-Major) to 5.2 tokens, making onset misalignment *representationally impossible by construction* rather than something the model must learn to avoid through self-attention. The practical consequence—verified by perceptual evaluation (GT = 3.92, matched to the Human Anchor at $p = 0.93$)—is that the encoding topology delivers professional-grade rhythmic cohesion unconditionally, independent of model scale or training data volume.

Limitations

Several limitations remain open for future investigation. First, melodic diversity within extended single-context windows is constrained by the 2048-token causal boundary; while Temporal Anchor Stitching alleviates macro-structural continuity, latent semantic drift in melodic motif development remains unresolved. Second, the current decoding temperature

provides a coarse handle on stochasticity; finer hierarchical temperature scheduling per instrument type (e.g., higher creativity on Chord tracks, lower on Kick to preserve groove rigidity) remains unimplemented. Third, our qualitative perceptual study, while encouraging, remains limited in participant scale and should be expanded to a formal double-blind MOS study with statistically sufficient participant pools in future work. Fourth, and most critically from an evaluation standpoint, this paper does not include a direct **generative quality baseline comparison** (PCE, PR, KL Divergence) between PulseFormer and a Track-Major model trained on identical data under identical compute budgets. Because the two paradigms employ fundamentally different tokenization schemes and decoding pipelines, a fair generative comparison requires a shared evaluation harness that decouples token format from MIDI output quality—engineering work beyond the scope of this paper. Our quantitative contribution is therefore best understood as a *structural* baseline (Section V-C), which provides a geometric proof of the synchronization advantage, rather than a full generative quality benchmark. We acknowledge this as a significant gap and commit to addressing it in subsequent work.

Future Work

Several compelling research directions emerge from this work. First, combining Time-Major interleaving with CP-style intra-note compression [7] could yield sequence lengths reduced by an additional $4\times$ factor, enabling single-context full-song generation without stitching. Second, integrating neural audio rendering (e.g., RAVE [11] or DiffWave) downstream of the MIDI decoder would enable end-to-end waveform synthesis while retaining the structural steerability advantages of symbolic generation. Third, extending the Energy conditioning framework to support continuous real-valued energy signals rather than discrete 9-level tags would enable smoother, more expressive macro-dynamic sculpting analogous to automation curves in professional DAW workflows. Fourth, constructing a **unified generative evaluation harness** that operates at the MIDI-object level (rather than at the token level) would enable a fair PCE/PR/KL comparison between PulseFormer and Track-Major baselines (e.g., MMM [4]) trained under identical conditions—resolving the most critical evaluation gap identified in the current work.

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